

TRIPLE POWER SUPPLY From USB

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Nowadays, most portable electronic gadgets are powered by USB adaptors. These adaptors have 5V, 1A to 5V, 2.1A output, which is enough to power low-power circuits of different voltages while testing a prototype. This is done either by using charge pumps or suitable converters. Presented here is the circuit to derive three voltages (3.3V, 5V and 12V) from a USB adaptor.

Circuit and working

As shown in Fig. 1, the circuit is built around low-dropout voltage regulator LP2985-3.3 (IC1), monolithic DC-DC converter MC34063A (IC2) a Schottky diode 1N5819 (D1), 22µH inductor (L1) and a few other components.

Here, MC34063A is configured in boost converter mode. IC2, inductor

L1 and diode D1 form the boost converter to convert 5V into 12V. Any readily available inductor can be used in place of L1. Resistors R1 and R2, along with 1.25V reference voltage of IC2, are used to set the output voltage using the relationship given below:

$$V_{out} = 1.25[1 + (R2/R1)]$$

Resistor R4 turns off the switching cycle when the peak current of the internal switch goes beyond the limit.

Capacitor C6 smoothen the ripple produced during every switching cycle. IC1, a fixed-output and low-dropout regulator, converts 5V into 3.3V. Capacitor C2 bypasses ripples to ground. Capacitors C3 and C5 act as buffers. Thus, all three voltages (3.3V, 5V and 12V) are available at CON2 to power external circuits.

PARTS LIST

Semiconductors:

- IC1 - LP2985-3.3 voltage regulator
- IC2 - MC34063A DC-DC converter
- D1 - 1N5819 Schottky diode

Resistors (all 1/4-watt, 5% carbon):

- R1 - 3.9-kilo-ohm
- R2 - 33-kilo-ohm
- R3 - 180-ohm
- R4 - 0.2-ohm

Capacitors:

- C1 - 1µF, 16V electrolytic
- C2 - 0.01µF ceramic disk
- C3 - 2.2µF, 16V electrolytic
- C4 - 330pF ceramic disk
- C5 - 470µF, 16V electrolytic
- C6 - 330µF, 35V electrolytic

Miscellaneous:

- CON1 - USB-A type connector
- CON2 - 4-pin berg strip male connector
- L1 - 22µH inductor
- USB - USB cable
- 2.5mm header female/female jumper wires

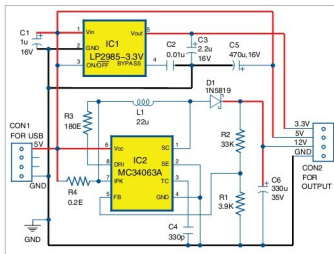


Fig. 1: Circuit diagram of triple power supply from USB

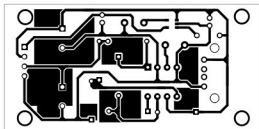


Fig. 2: Actual-size PCB layout of the triple power supply

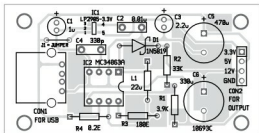


Fig. 3: Components layout for the PCB

Construction and testing

An actual-size PCB layout of the triple power supply from USB is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on the PCB, enclose it in a suitable plastic box with CON1 fixed at the rear side and CON2 at the front side of the box.

IC1 should be placed on solder side of the board. Use USB-A type connector as CON1.

After assembling the PCB, connect the USB to a laptop. Now your power supply circuit is ready to use.

Note. 1. The maximum current for 3.3V and 12V depends on the current provided by USB port.

2. Since 5V is directly available from the USB, keep the overall current of this circuit below 20-25 per cent of the maximum current of the USB from which the circuit is powered. **EFY**



A. Samiuddin is B.Tech in electrical and electronics engineering. His interests include LED lighting, power electronics, microcontrollers and Arduino programming